

MISSION 1:

INHALATION

(MENARIK NAFAS)

Teacher, what happens to the muscles between our ribs when we breathe in?

Great question! Let's look at the first thing that happens.

1

When we inhale, the **EXTERNAL INTERCOSTAL MUSCLES CONTRACT...**

External intercostal muscles **CONTRACT**

2

...while the **INTERNAL INTERCOSTAL MUSCLES RELAX.**

Ohh... so one set contracts and the other set relaxes!

Internal intercostal muscles **RELAX**

3

Exactly! The external intercostal muscles pull the ribs up and out, and the internal intercostal muscles relax to allow that movement.

This helps to **enlarge the space in the chest.**

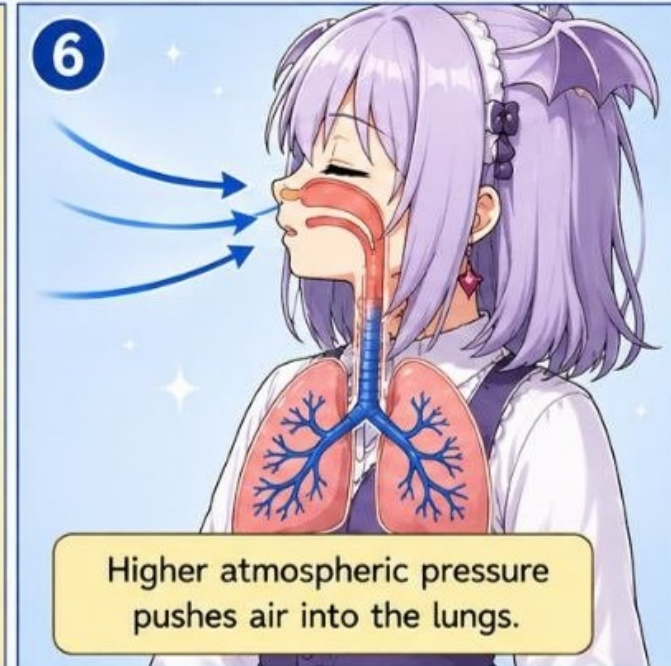
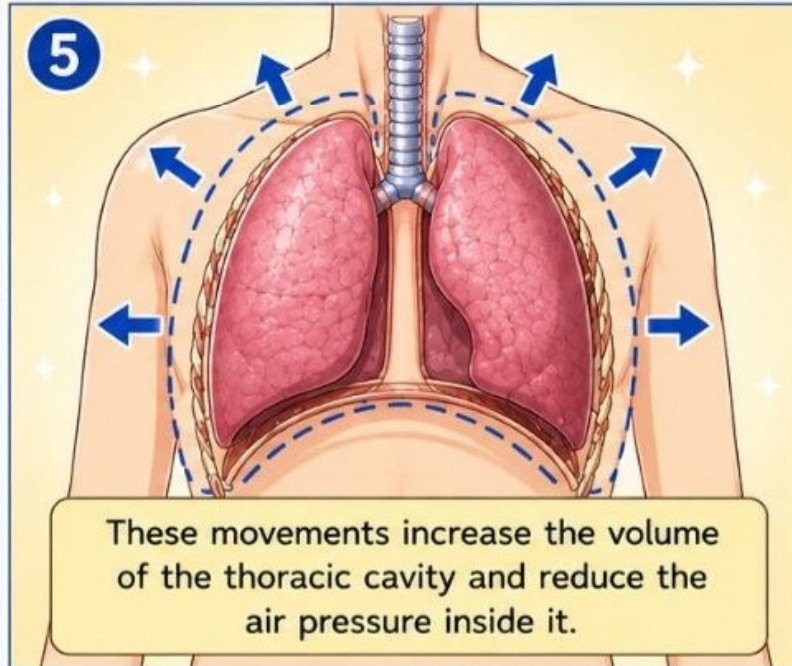
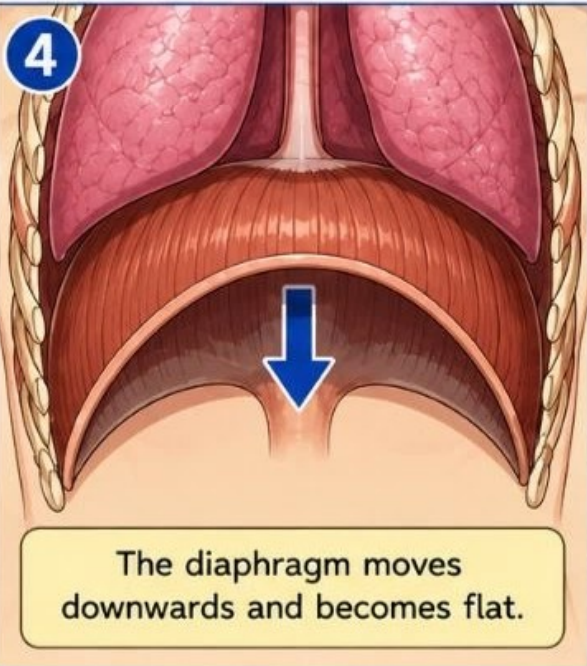
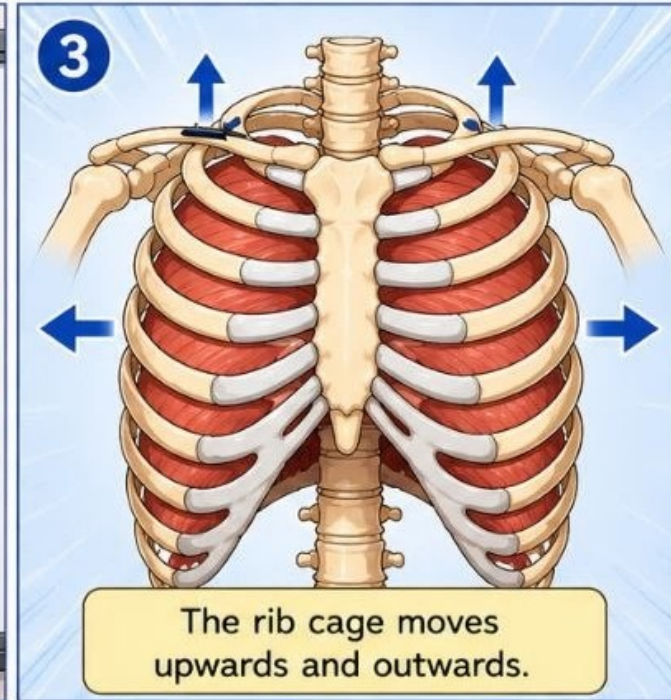
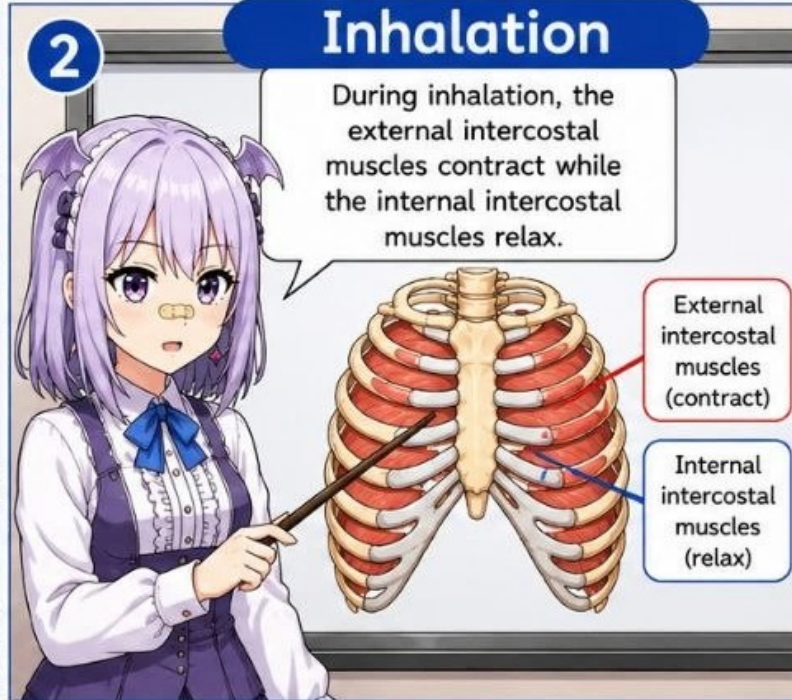
4

So to recap for Step 1:

- External intercostal muscles **CONTRACT**
- Internal intercostal muscles **RELAX**

Got it! First, the muscles prepare by contracting and relaxing!

STEP 1 COMPLETE!
ON TO THE NEXT CHANGE!



EXHALATION (BREATHE OUT)

1

So, what happens when we breathe out?

2

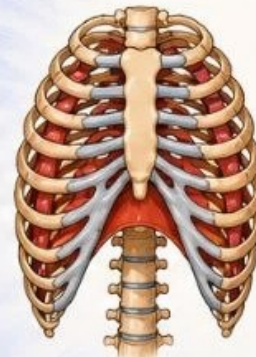
During exhalation, the external intercostal muscles relax while the internal intercostal muscles contract.

External intercostal muscles RELAX

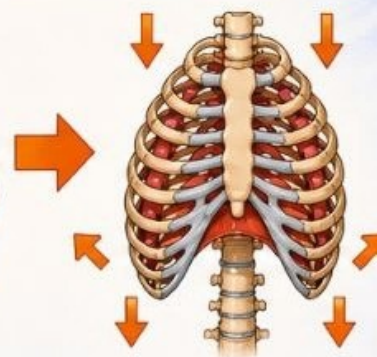
Internal intercostal muscles CONTRACT

3

The rib cage moves downwards and inwards.



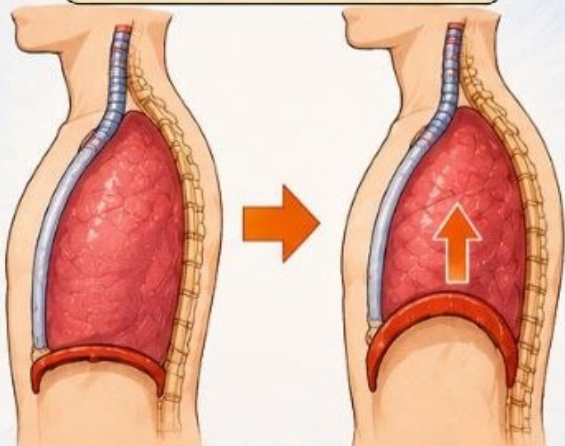
BEFORE (INHALATION)



DURING EXHALATION

4

The diaphragm curves upwards to form a dome.



BEFORE (INHALATION)

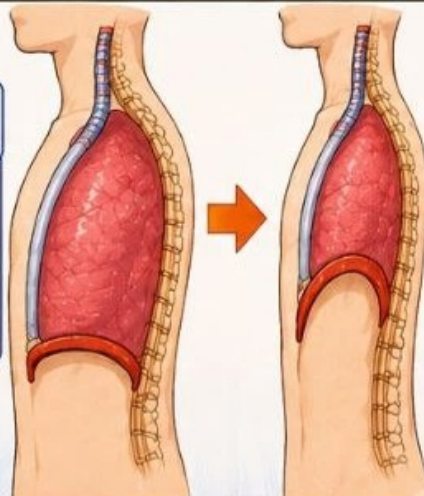
DURING EXHALATION

5

The volume of the thoracic cavity decreases and the air pressure inside it increases.

BEFORE (INHALATION)

Volume LARGE
Pressure LOW



DURING EXHALATION

Volume SMALL
Pressure HIGH

6

Now I understand—air is pushed out of the lungs!



1 REVIEW: INHALATION vs EXHALATION

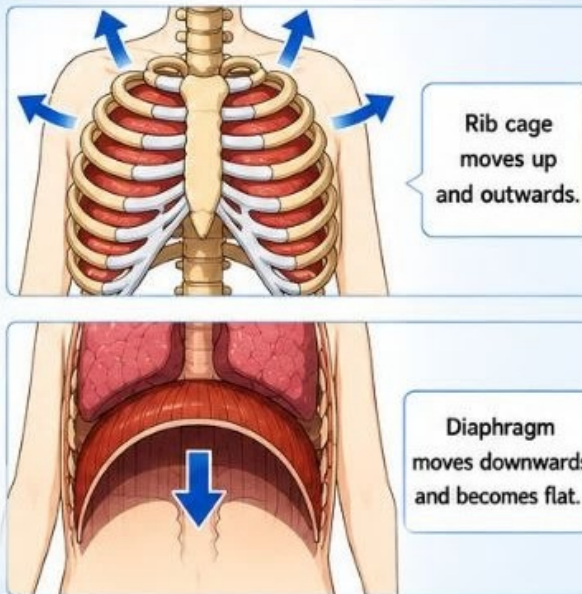
INHALATION

FEATURE

EXHALATION

Rib cage moves up and out		RIB CAGE		Rib cage moves down and in
Diaphragm moves down and becomes flat		DIAPHRAGM		Diaphragm moves up and becomes dome-shaped
Thoracic cavity volume increases		THORACIC CAVITY VOLUME		Thoracic cavity volume decreases
Air pressure decreases		AIR PRESSURE		Air pressure increases
Air enters the lungs		AIR MOVEMENT		Air leaves the lungs

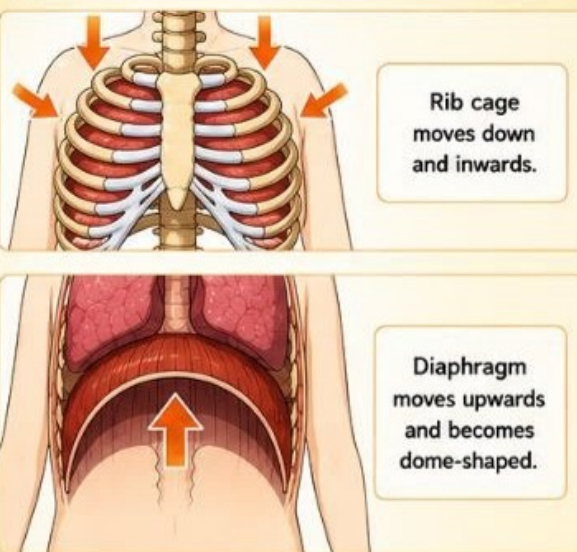
2 INHALATION: SUMMARY



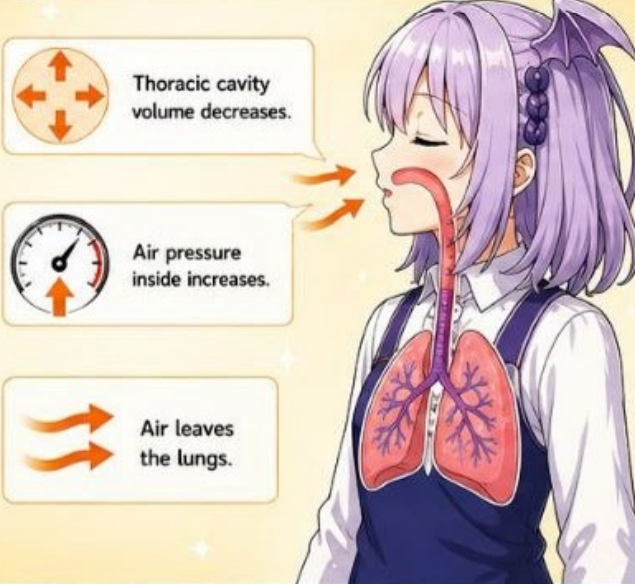
3 INHALATION: RESULT



4 EXHALATION: SUMMARY



5 EXHALATION: RESULT



Every breath keeps
us **alive!** 



BREATHE IN

- The thoracic cavity expands and air **enters**.



BREATHE OUT

- The thoracic cavity becomes **smaller** and air **leaves**.



MORAL VALUES

-  **APPRECIATION OF HEALTH** – Be grateful for the ability to breathe and stay healthy.
-  **CURIOSITY** – Keep asking questions and keep learning.
-  **RESPONSIBILITY** – Take care of your body and make healthy choices.
-  **HEALTHY LIVING** – Exercise regularly, eat well, and keep your lungs strong. 

Take care of your lungs—
every breath is **precious**. 

Let's breathe well,
live well, and be well! 

